TTG continuation: partial performance study

LittleX and IssueDesk | all six policies | n=3 independent workflows per cell

48 fresh workflow trials

12 excluded warmups + 36 measured workflows

Primary metric: actual SQL send -> ReadyForQuery completion wall time.

Summed DB calls, interval union, SQL data/control counts and real PID overlap.

Identical seeded workflows and independent response/state correctness guards.

Scope is two applications; the full application and paper tiers remain separate.

LittleX: DB time and SQL counts

Median [IQR]; three fresh measured workflows for every row

Policy	DB sum ms [IQR]	DB union ms [IQR]	SQL total/data/control/other	FG/BG distinct backend overlap ms
Off	39.859 [1.236]	39.859 [1.236]	153/90/63/0	0.000
Legacy TTG / thread	116.583 [1.556]	116.400 [1.532]	144/74/70/0	0.141
Always / thread	50.362 [8.214]	43.831 [7.562]	167/90/77/0	0.000
Always / process	49.514 [2.891]	49.514 [2.891]	159/90/69/0	0.000
Expected benefit / thread	47.846 [2.153]	42.873 [1.440]	165/90/75/0	0.000
Expected benefit / process	50.053 [1.206]	50.053 [1.206]	159/90/69/0	0.000

Overlap column = foreground/background protocol intervals on distinct PostgreSQL backends.

SQL counts are dispatches; actual completion cycles and all raw samples are in the data export.

IssueDesk: DB time and SQL counts

Median [IQR]; three fresh measured workflows for every row

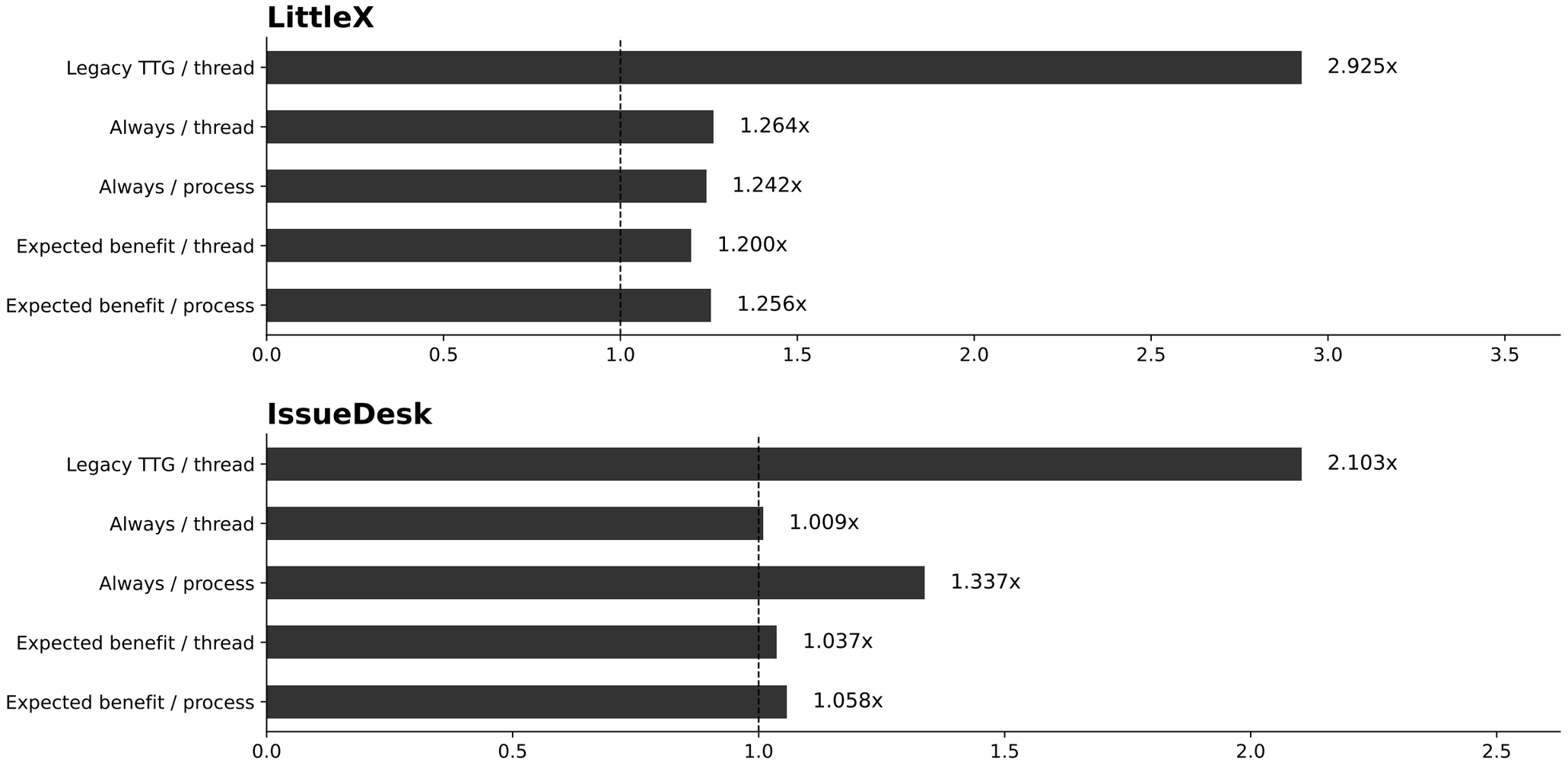
Policy	DB sum ms [IQR]	DB union ms [IQR]	SQL total/data/control/other	FG/BG distinct backend overlap ms
Off	140.773 [11.417]	140.773 [11.417]	555/366/189/0	0.000
Legacy TTG / thread	296.063 [16.585]	293.819 [16.541]	569/354/215/0	3.043
Always / thread	142.029 [2.018]	142.029 [2.018]	563/366/197/0	0.000
Always / process	188.234 [24.292]	188.234 [24.292]	559/366/193/0	0.000
Expected benefit / thread	145.973 [5.113]	145.973 [5.113]	561/366/195/0	0.000
Expected benefit / process	148.872 [0.697]	148.872 [0.697]	558/366/192/0	0.000

Overlap column = foreground/background protocol intervals on distinct PostgreSQL backends.

SQL counts are dispatches; actual completion cycles and all raw samples are in the data export.

Relative summed DB-call time

Ratio of workflow medians to off; <1 lower, >1 higher. Dispersion is shown in the timing tables.



Measurement and provenance

Actual DB evidence; independent correctness and exact source binding

Protocol completion wall includes row decoding/client progress, not server CPU alone.

Sum/union gaps cover any overlapping calls; FG/BG overlap is a separate metric.

Late prefetch SQL remains attributed to its original business request.

Process startup/IPC remains in raw model/lifecycle evidence; no guessed subtraction.

SQL logging is off in measured trials; independent reconciliation uses separate controls.

Instrumented HTTP is auxiliary. No clean e2e or application-CPU conclusion is claimed.

Accepted export SHA256: 17922e61bcf4ff4ca6814a1b23cd758d0ac371d979f1537ab95dc802233c2f4b

Main authorization SHA256: d1fc7443300f79ce5bc2185f900182466d60bed336906d99c54540c585d1dc02